

Connection admission control in ATM networks

P B Key

Connection admission control is of vital importance for ATM networks. Any control strategy must balance network and user demands, satisfying user quality of service requirements while at the same time allowing the network to run efficiently. This paper provides a layered solution to the problem, with particular emphasis on real time services.

A time-scale decomposition is given which aids understanding and analysis — buffers in the network are dimensioned to cope with variation on a short time-scale, and related to the way connections are policed, while on a longer time-scale the number of connections is limited to ensure the available bandwidth is not exceeded. The simplest way to limit connections is to allocate each connection its peak bandwidth. However, it is possible to do much better than this and fit much more on to the network; this paper shows how it is possible to safely statistically multiplex connections by allocating an effective bandwidth to a connection. An adaptive call admission control is presented which allows the effective bandwidths to be estimated in real time, with no further information required from the user, and which explicitly treats the trade-off between cell loss and call rejection.

1. Introduction

Connection admission control says how and when calls should be admitted to a network. ATM is connection-oriented, and when a connection arrives at the network boundary, the network has to decide whether to admit the call or not. At this stage, a contract negotiation occurs — the user declares a set of requirements to the network, and the network agrees to carry the call at a given quality of service. Renegotiation can take place if the network is unable to satisfy the user's initial demands. In practice this whole process may be transparent to the user, with terminal equipment communicating with the network automatically.

In a network, call admission control, routing, congestion control and tariffing are all intertwined. In a lightly loaded network all calls will be accepted, whereas when the load increases, certain calls will be rejected. Some might be rejected because the network is unable to carry them without compromising the quality of service of other calls (lack of capacity), and others rejected because the network prefers to wait for a higher revenue call to arrive. Whether the network can accept a new connection depends on the flexibility of the routing strategy — there may be spare capacity in unused parts of the network, whereas preferences amongst call types depend partly on tariffing issues.

This paper largely side-steps issues of routing, and only mentions tariffs in passing. Instead it addresses the basic problem of deciding whether there is enough capacity in the network to admit a call of a particular type. Routing and tariffing can be thought of as second-order effects

which can be looked at once the basic problem has been identified and answered.

To some, an ATM network conjures up visions of the M25, where congestion can arise spontaneously, and once it has started, takes a long time to go away. Traffic either seems to be flowing at 70 mph or not moving at all! There are some similarities between road networks and ATM networks — cells or cars traverse a network, and have to wait at various nodes/junctions in the network, with information travelling at different speeds in different parts of the network. However, there are some important differences:

- ATM networks have access control that allows excess demands to be temporarily refused — this is a key factor, which allows the network to be run at sensible loadings, rather than at critical loadings where the network can flip into congestion,
- cells are in the network for a short time,
- connections are policed to ensure that they do not violate their contract.

With these differences, a network can be run in an analogous way to the telephony network — call attempts are rarely refused service, and once accepted are routed to their destination, possibly using an intelligent or dynamic routing scheme, and the call has a guaranteed bandwidth until the connection is released. Congestion when it does

occur is short lived and confined; once the cause of congestion abates, the network reverts almost immediately to a lightly loaded state. There is no persistent rush-hour effect.

It is shown how it is possible to run broadband networks in such a way; but firstly, the paper looks at two different views of networks and services.

2. Views of the world

An ATM network can be run in several ways — in one, all demands are accepted when they are offered to the network, and the network exerts back-pressure or flow control when the network approaches congestion. Controls throttle back the load offered to the network by telling or forcing users to slow down. This closed-loop control is the approach of traditional packet-switched networks. Recently, this approach has gained ground in the ATM Forum, under the so-called ‘available bit rate’ heading, which is discussed by Adams and Smith [1].

This is also an Internet view of the world — users share bandwidth, and when there is contention for resource, users expect ‘fair’ treatment, but there are no guarantees. When the network is lightly loaded, users can have as much bandwidth as they require. As the load increases, spare capacity will be exhausted. A new connection will still be admitted, but now the bandwidth of some existing connections will be decreased. Roughly speaking, this ‘best effort’ service is appropriate to users who expect some bandwidth, where some is better than none. More formally, Shenker [2] shows that this strategy is optimal if the utility or value that users place on the bandwidth is a concave function. A classic example is file transfer, where typical transport protocols adapt to the bandwidth they receive.

A simpler method of control is open-loop feedback — sources are allowed to transmit on to the network, and once accepted are regulated in a set way, independent of the state of the network. For example, in telephony or circuit-switched networks, the only constraint is on the capacity of the connection, and that capacity or bandwidth (and quality of service) is guaranteed for the life of the connection, with the price of the service reflecting these guarantees. The controls within the network are much simpler, though now some care has to be taken over admitting calls, anticipating the future consequences to other calls already admitted. This is the approach discussed in this paper.

An appropriate service model is where users require some definite amount of bandwidth (or more generally, where the utility placed on bandwidth is convex [2]). A natural example for this is real time services. This situation is very different from the Internet model. It is not worthwhile reducing the bandwidth allocation of a connection below its maximum desired value — so if all the

capacity is allocated, a new connection should be rejected. This also implies networks have to be dimensioned correctly and may have to run with lower utilisations.

ATM is fundamentally connection-oriented, and the contract negotiation discussed earlier points in this direction. Yet, at the same time, ATM is a fast-packet technology, and ATM switches involve buffering. How can such a network behave in a similar fashion to a circuit-switched network? One answer is that ‘small’ buffers can be chosen in ATM (much smaller than a round-trip delay time), and connections can be allocated an ‘effective bandwidth’, enabling the network to look as if it were a circuit-switched network. These concepts are spelled out in the following sections.

3. Buffers and timescales

Hui [3] was the first to introduce the notion of time-scales — connections last for a ‘long’ period of time (perhaps measured in seconds or minutes); this is the call-time scale. Looking more closely, perhaps on a millisecond time-scale, it can be seen that a call is made up of bursts of activity interspersed with inactivity, which is called the burst scale. Moving in closer still, to a microsecond or cell scale, each burst will comprise a number of cells, each sent at the line rate, with intervening gaps (see Fig 1).

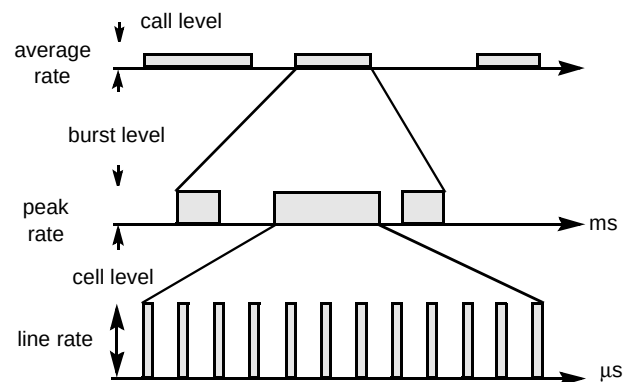


Fig 1 Time-scales.

3.1 Performance parameters

Different performance metrics apply at the different levels. At the call or connection level, it is the ‘grade of service’ that is of interest, the fraction of calls that are denied access because of lack of network resource. A figure of 1 in 1000 is currently typical of telephony networks. Appropriate traffic descriptors are the call-arrival process and statistics of call-holding time.

At the cell or burst level, cells can be lost or delayed. Traffic descriptors at this level relate to burst-length

distributions, inter-burst distributions and within-burst cell-arrival statistics.

One of the key performance measures is the cell loss ratio (CLR), defined as ratio of cells lost to cells offered. Some of the services that can be carried on a broadband network are very sensitive to loss, and CLRs of 1 in 10^9 or higher are talked of, the same ratio as 1 second in 32 years. To put this in context, this is the same order of magnitude as would be induced by bit errors from an optical line system. It corresponds to one lost cell every 40 min on a 155 Mbit/s line system, one lost cell every 4.5 days for a 1 Mbit/s connection. Other services, such as voice or certain video coding schemes can tolerate a much lower quality of service, several orders of magnitude lower than this (e.g. a CLR of 1 in 10^3 for voice).

Delay related measures are the other main performance parameters, comprising quantities such as mean transfer delay and cell delay variation (CDV), which is discussed below.

These are all ATM layer performance parameters and need to be translated into application level performance, e.g. a video application is interested in degradation of frames and frame loss rather than cell loss.

3.2 Cell scale buffers

A simple and appropriate model of an ATM switch is a non-blocking crossconnect with output buffering. The larger the output buffer, the larger the queuing delay, which is bad for real time services, whereas buffers can store excess load, enabling networks to be run more efficiently. How large should the buffers be? They are intimately linked with the amount of CDV tolerance allowed, which is discussed below, together with how the networks are run. For open-loop control, only cell scale buffers are needed, not burst scale ones.

Cell scale buffers handle the phasing problems associated with multiplexing traffic (i.e. simultaneous cell arrivals from different traffic streams) given that the load is less than the capacity. For instance, if there are two constant bit rate traffic streams, each half the link bit rate, contending for a buffer, then these might be in phase, requiring that one cell be buffered. Burst scale effects occur when the arrival rate exceeds the service rate, so burst scale buffers store this excess demand.

Figure 2 illustrates the queuing behaviour of a number of bursty sources multiplexed together in a simple first-in-first-out queue. As the buffer size is increased, the probability that the buffer overflows decreases exponentially fast, while buffer space for the cell component is still being allocated. Beyond a certain point

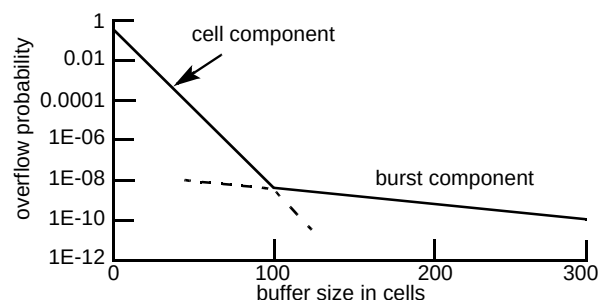


Fig 2 Burst and cell scale effects.

(the 'knee'), the behaviour changes sharply, and the decrease is much more gentle. This is now the region of burst scale queuing, where a large amount of extra buffering is needed to decrease the buffer overflow probability. Moreover, the cell-scale component is remarkably insensitive to the statistics of the arrival processes, whereas the burst scale is very sensitive to burst distributions and burst lengths, which is another reason for preferring cell scale buffering.

The cell scale buffering required depends on the load. With a moderate load (0.75) and constant bit rate sources, a buffer of size 50 is sufficient for negligible cell loss. At high load (0.95), a buffer of size 200 is sufficient for random traffic (150 for CBR sources) (see Roberts [4] for details of the queuing models used).

3.3 Time-scale separation

In what follows it is assumed:

- cell scale buffers are large enough to cope with CDV and simultaneous arrivals,
- connection admission control ensures that burst-level congestion (when the demand exceeds capacity) only occurs with small probability, sufficiently small to meet the cell loss constraints,
- network dimensioning is sufficient to ensure suitable call-level blocking (grade of service).

These time-scales are treated as separate, enabling the problem to be decomposed. The call time-scale is much longer than cell scale, and burst scale variation is seen by the call process as being in equilibrium.

4. Peak rate allocation

The simplest way to run an ATM network is to allocate each connection its peak bandwidth. If transmission has a certain capacity R , such as 150 Mbit/s, and a connection requires bandwidth r_i , then the sum of the allocated band-

widths must be less than the capacity, where r_i and R are measured using the same units, such as cells per second. This constraint can apply at an output port of a switch, or when virtual connections are added to a path. However, ATM switches have buffering to resolve contention, so they have to be run at some utilisation (ρ) less than 1 to avoid losing cells or the queue becoming unstable (recall the M25 analogy). For example, the switch might run at a utilisation of say 80%, $\rho = 0.8$, in which case the constraint is:

$$\sum_i r_i \leq \rho R \quad \text{..... (1)}$$

that is:

$$\sum \text{peak rates} \leq \rho \times \text{capacity}.$$

More generally, the input stream may not be precisely specified by its peak rate. This might be because the customer is connected to some multiplex equipment which distorts the user's traffic, or perhaps the information arises from a data or video source which naturally sends out data in clumps. The peak rate of such sources could be defined to be the reciprocal of the minimum inter-arrival times T of cells as seen by the UNI (user/network interface); however, this is unduly pessimistic, could cause the full link to be allocated to most connections, and is not the way standards define the interface. Instead policing and the variation allowed to the customer are bundled together — the customer gives to the network two parameters, a peak rate, and a cell-delay variation tolerance τ which quantifies the 'jitter' [5].

4.1 Cell delay variation

It is helpful to think of this in terms of one implementation of a policer, the so-called leaky bucket. The bucket is a token bank of maximum size $T + \tau$; if an arriving cell finds the bucket bigger than τ , then the cell is discarded (or tagged), otherwise the cell is accepted, and a token of size T is thrown in the bucket. The increment T is defined as the reciprocal of the peak rate and hence is the minimum contracted inter-arrival time. The bucket leaks out at unit rate, and the cell-delay tolerance τ measures the time it takes to empty a full bucket if no more cells arrive. Equivalently, there can be tokens of unit size, with a leak rate of the peak rate. Therefore the 'peak rate' specified in the contract is really the leak rate of a leaky bucket, and measures the maximum arrival rate over a period of time. A couple of points are worth mentioning:

- the leaky bucket is really a virtual leaky bucket — cells are not delayed, merely accepted or discarded (or tagged),

- cells could be delayed, only being allowed to leak out at the peak rate, in which case the leaky bucket becomes a spacer (traffic shaper) and reshapes the cell stream,
- the leak rate of the bucket should be slightly greater than the claimed rate of the source.

This last point is to avoid synchronisation problems, and also ensures stability of the policer.

4.2 CDV and the first multiplexer

Leaving aside for the moment the problem of how to choose the tolerance, suppose that there are a number of sources, each policed by leaky buckets. Let the leak rate and bucket depths of these policers be r_i cells per second and b_i cells, so the CDV tolerance is $\tau_i = (b_i - 1)/r_i$. If these sources are multiplexed together, perhaps at the output port of the first switch, they will compete for the buffer capacity B . Since the worst case output from the policer is a group of b back-to-back cells¹, it is intuitive that no cells should be lost, provided that the sum of the bucket depths of the policers is less than the buffer capacity of the switch, as illustrated in Fig 3.

This is indeed the case, as can be simply shown, using a bounding argument first used by Cruz [6]. In an interval of time $(-t, 0)$ the number of cells allowed through by leaky bucket i is bounded above by $b_i + tr_i$; hence the number of cells arriving at the buffer in an interval of length t is bounded above by $\sum_i b_i + t \sum_i r_i$. Using equation (1) it then

follows that no cells will be lost in the shared buffer provided that:

$$\sum_i b_i \leq B \quad \text{..... (2)}$$

which just says that the sum of the leaky bucket depths should be less than the output buffer in the switch.

This constraint can be unduly pessimistic for small b . For instance if there are 1000 constant bit rate sources with no CDV ($b = 1$), then this cautious approach would say a 1000-cell buffer was needed. Whereas if a simple deterministic queuing model is used and the sources are assumed to be randomly phased, only a 90-cell buffer is needed to give negligible loss. Indeed, as mentioned in section 3.2, a 150-cell buffer is sufficient to cope with any number of multiplexed sources of size 1, and a buffer of size $150b$ is sufficient for sources which each transmit b back-to-back cells.

¹ In fact the number of back-to-back cells is bounded by $b/(1 - r/R)$.

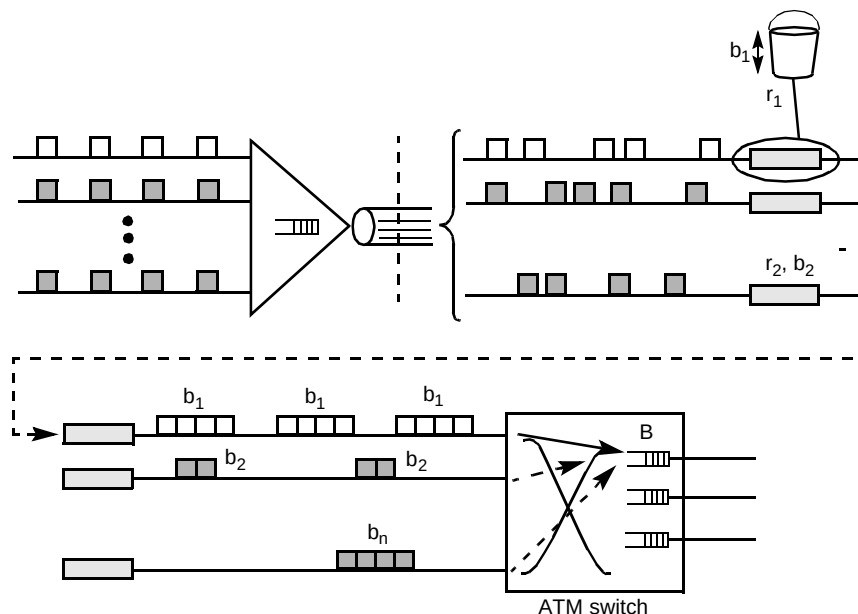


Fig 3 CDV and buffer sizes.

In summary, there are two conditions to be met — a utilisation constraint and a buffer constraint. In fact, the two constraints can be replaced by a single constraint; suppose the bucket depths are chosen so that:

$$\frac{b_i}{r_i} \leq \frac{B}{R} \quad \text{..... (3)}$$

which says that the time to empty the leaky buckets is bounded above by the time taken to empty the buffer. Then the buffer equation (2) is implied by the utilisation constraint (1). This provides a ‘natural’ bound for CDV; if a user is policed by a leaky bucket with parameter r_i then a natural limit for CDV is to use a bucket depth:

$$b_i = r_i \frac{B}{R} \quad \text{..... (4)}$$

corresponding to a CDV tolerance bounded above by B/R . This clearly shows the effect of buffer sizes in the switches — the larger the buffer in the switch, the larger the CDV that can be tolerated. For example, for a 155 Mbit/s network, a 100-cell buffer switch allows some 0.27 ms

CDV, 200-cell buffer 0.54 ms, and so on. It should be noted that slightly more than this can be allocated and the utilisation ρ can be taken into account, in which case the bucket depths (equation (4)) are multiplied by $1/\rho$.

Using this approach, only the peak rate need be known, and then policed accordingly. What happens if a user needs a larger tolerance than this ‘natural’ amount? In this case, it is as if the user actually requires a large peak rate. (Explicitly, if the user wants a tolerance τ_i larger than the natural amount B/R , the demand can be treated as if the source requested a peak rate $r_i \times \tau_i R/B$.)

4.3 User values for CDV

Access multiplexer

The magnitude of the CDV that a user requires depends upon the application generating the traffic, and also how it reaches the network.

How large a tolerance does a user require? An end-station might be connected to a (private) local area network which is then connected to a public network via a multiplexer, causing the network to see a distorted version of the original traffic (see Fig 3). Although the access multiplexer could be arbitrarily bad, a ‘first-in-first-out’ queue can be taken as a reference case. In this case, the ‘natural’ limit that has been given is enough to cater for the distortion that constant bit rate streams see, provided that the stream is policed at a higher rate. The exact increase depends on the buffer size and quality of service standard

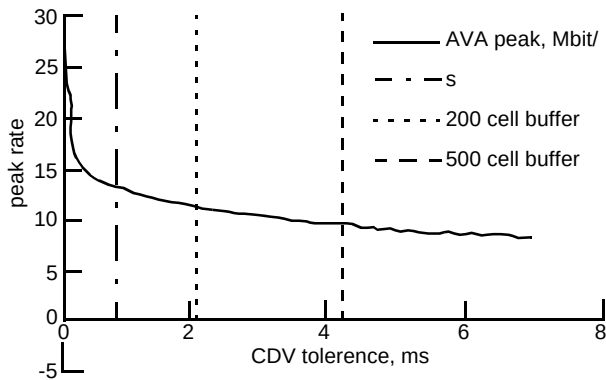


Fig 4 Peak rate and CDV tolerance for an AVA camera.

[7]. For instance if there is a 200-cell buffer switch, the required increase is less than 5%.

An optimal user choice

Alternatively, a user might choose the 'best' peak rate and tolerance to minimise some incurred costs. For a given offered traffic, there is potentially a range of peak rates and τ_i values which would allow the offered traffic to pass a leaky bucket at a fixed quality of service.

Figure 4 shows the peak rate and CDV tolerance which allows the output from a Fore Systems AVA 200 video camera to pass unscathed through a leaky bucket. The camera in question uses JPEG compression on 'tiles' of 24-bit colour pels, and converts the image into ATM cells using AAL5. In this example the camera generated 25 frames per second. Figure 4 was generated from a particular 25 frames per second trace of a TV transmission [8] with the camera's spacer disabled, allowing back-to-back cells to be transmitted, giving a potential peak rate of 100 Mbit/s and a mean of 2.74 Mbit/s. The inter-arrival times from the trace were used to calculate the bucket depths and leak rates.

From the diagram, it is clear that the user could choose to offer a peak rate of 27.5 Mbit/s to the network with minimal CDV tolerance, or a peak rate of 7.9 Mbit/s and a CDV tolerance of 7 ms. Which value should the user choose? From the network's point of view, the larger the peak rate, the more resource that is consumed; if this is reflected in tariffs which charge on a per-cell basis, with the per-cell rate dependent on the peak rate, then the user would want to choose the smallest peak rate possible, subject to the CDV tolerance allowed by the network being large enough.

The vertical lines in the figure represent the 'natural' CDV tolerance described in section 4.2 for different size switch buffers, namely those of 200, 500 and 1000 cells, with a line rate of 100 Mbit/s. For the a 200-cell buffer, the natural limit of 0.84 ms CDV tolerance implies a peak rate of 13 Mbit/s for the AVA camera; if the switch has 500 or 1000 cell buffers then the source could use a peak of 11 or

9 Mbit/s respectively. This illustrates well the link between buffer sizes in the switch and allowable CDV.

4.4 The network problem

As a traffic stream goes through a network, its cells mingle with other traffic streams, and the stream is distorted. Two scenarios could be envisaged — in the first, traffic which starts off with a small CDV gradually accumulates CDV as it progresses through the network, and clumps of cells coalesce to form even bigger clumps; in the second scenario, clumps of cells are gradually broken up as they pass through the network, the interfering streams acting to disperse clumps of cells, so the number of back-to-back cells decreases the further into the network it progresses. Which of these two effects dominate?

In fact, the answer is that they both apply in different circumstances. Constant bit rate sources become more distorted as they traverse a network, while extremely large clumps of cells are likely to be split up.

In an elegant piece of theory, van den Berg and Resing [9] provided some insight into this. Consider a sequence of heavily loaded queues in series, and a reference stream. At each queue, the reference stream encounters independent background traffic which mixes with the reference stream before going elsewhere. If the queues are heavily loaded, then at the exit of the n th queue the distance between successive cells of the reference connection is just the time taken to serve the number of cells from the background traffic which arrive between the inter-exit times of the $n-1$ th queue. This is a simple branching process, which enables all the statistics of the reference stream to be calculated.

It turns out that all that matters is the background traffic stream. If there are enough queues, the output stream will tend to a limit, which depends only on the background cross-traffic stream. The higher the bit rate of the connection, the quicker the convergence. For random cross traffic, this limit has a variance which is half that of a Poisson stream, thus quantifying which streams are smoothed, and which get worse.

- Traffic which is smoother than this limit increases in CDV as it passes through the network. In particular constant bit rate traffic increases in CDV as it passes through the network, but not without bound — the amount can easily be calculated [10]. Therefore, if a connection needs to be policed later on in a network, perhaps at an intercarrier interface, the parameters have to be looser than at the entrance to the network.
- Streams which are more bursty than the limit are smoothed as they pass through the network.

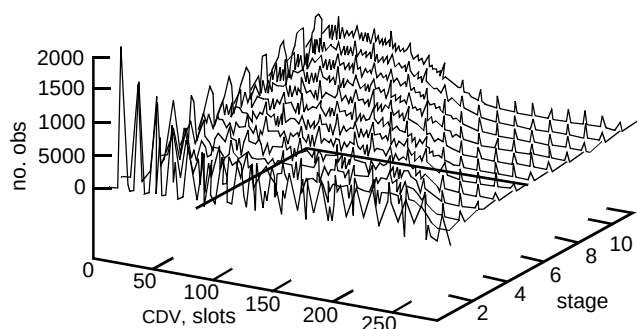


Fig 5 CDV for a stream passing through a series of queues.

This theory is borne out in practice as shown in Fig 5. This plots the actual CDV experienced on the Fairisle ATM testbed in Cambridge when a reference stream of 20 back-to-back cells passes through a series of queues, where CDV is measured in slots. At each queue, the reference stream is mixed with 'crossing' background traffic [8], and each queue is heavily loaded (0.95 load). The spikes are the peaks caused by the back-to-back cells, which are gradually smoothed out as they go through the network.

Recall that the buffer sizing is based on the worst-case output from a leaky bucket, namely blocks of back-to-back cells. However, it is known that such blocks of cells become dispersed, and therefore at a later queue in the network the combined effects will be less onerous than at the first queue. Consequently, CDV only needs to be considered in terms of admission control at the entrance to the network.

5. Variable bit rate traffic and statistical multiplexing

PPeak rate allocation is inefficient for variable bit rate sources. In this section it will be shown how it is possible to statistically multiplex, thereby using the network more efficiently, while at the same time guaranteeing qualities of service. The approach is to use the decomposition hinted at earlier — buffer space is allocated to handle jitter (CDV), both that inherent in the source and that induced by interactions with other traffic, leaving capacity (bandwidth) allocation to meet quality of service requirements.

Previously each source was allocated its peak rate. Now this allocation needs to be reduced. To do this, traffic has to have a variable bit rate (not always transmitting at its peak rate), and several streams of traffic need to be mixed, so that non-coincidences of 'busy' and 'idle' periods of the sources can be exploited. Figure 6 shows that such multiplexing is possible, by depicting the bandwidth requirement of four sections of the *Star Wars* film (see Garrett and Willinger [11] for details of the coding). It should be noted how the bandwidth profiles are very different, and the whole trace has certain 'fractal' properties — but they can be multiplexed.

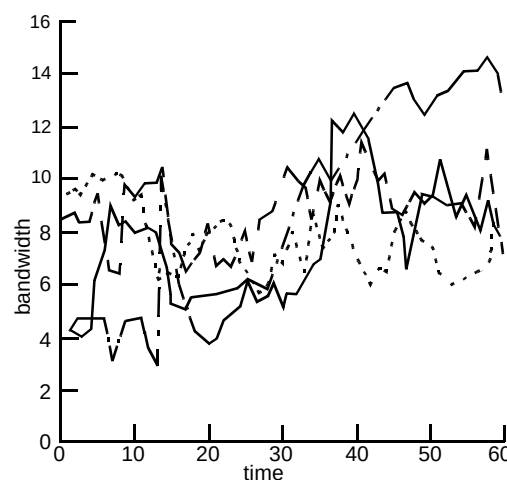


Fig 6 Bandwidth requirement for *Star Wars* traces.

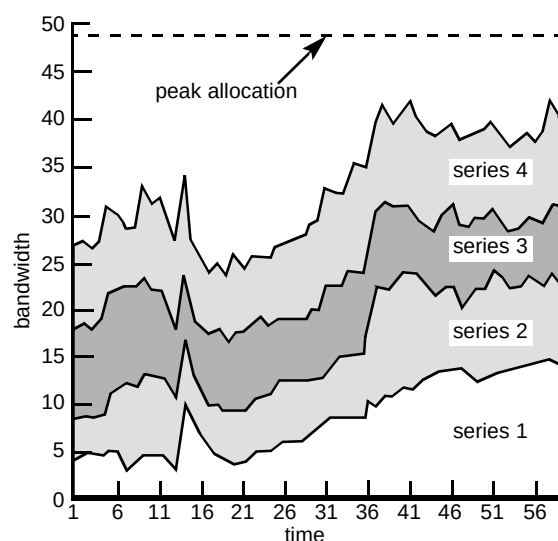


Fig 7 Multiplexing *Star Wars* traces.

Figure 7 shows how the traces fit together if they are multiplexed (corresponding to showing four different time-shifted versions of the film). The combined total bandwidth is less than the sum of the peak bandwidths. This assumes minimal buffering. If a modest amount of buffering is provided to smooth the sources, the gain is much larger [11].

In the light of the time-scale decomposition, only fluctuations of bandwidth on a coarse time-scale need be known, making sure that at any point in time the bandwidth required only exceeds the capacity with small probability (the quality of service constraint). On this 'burst-scale' it is as if there were no buffer at all. This guarantees erring on the side of caution, since some excess demand will be absorbed by buffers in the switch. To emphasise this point, the burst-scale fluctuations are on a longer time-scale than the time to empty a buffer, or equivalently the buffers are 'small'.

For a source with a given peak rate, the number of sources that can be fitted on to the link is known. If the effects of CDV are ignored, or the natural limit described in section 4.2 used, then the capacity needed to be allocated for all the sources is given by:

$$C = \rho R \quad \text{..... (5)}$$

using equation (1). It is possible, if desired, to use more sophisticated models for CDV, in which case the capacity C is related to the number of sources that can be multiplexed for a given buffer size and given CDV tolerance (see the remarks following equation (2)).

5.1 A simple model for sources

Suppose for the moment there is a single type of traffic, and a number, n , of independent sources of this type. Let $X_i(t)$ be the instantaneous bandwidth of source i , then the aggregate requirement of the sources is the load S defined by:

$$S_n(t) = \sum \text{bandwidths} = X_1(t) + \dots + X_n(t) \quad \text{.....(6)}$$

To meet given quality of service constraints, it is necessary for the load to exceed the capacity (C) with small probability. In fact, there is usually a requirement to bound the cell-loss ratio of a stream:

$$\text{cell-loss ratio} \leq 10^{-\gamma} \quad \text{..... (7)}$$

for some given number γ . For the aggregate stream (and hence an individual stream for this model), this is the ratio of the expected number of cells lost to those offered.

To progress, the bandwidth requirement needs to be modelled. The simplest way to do this is to use an on/off model — it is assumed that the source is either on at the peak rate, r , or else off:

$$X_i = \begin{cases} r & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases} \quad \text{..... (8)}$$

— see Fig 8.

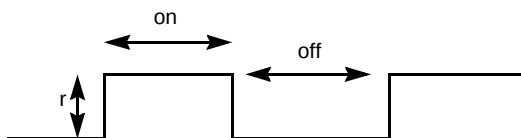


Fig 8 On/off burst-level source model.

The mean is rp . The burstiness, the peak-to-mean ratio, is $1/p$, and p is the activity of the source — the proportion of time it is active.

This model is clearly conservative (see Griffiths and Key [12]), and enables all the relevant quantities to be calculated easily. It should be noted that:

- the model is burst-scale and not cell-scale (cells are not generated with probability p , but bursts are),
- in the context of earlier sections, the peak rate of the source, r , is the negotiated peak rate, in other words the leak rate of the policing leaky bucket, and not the observed minimum cell inter-arrival time (this means it is time-scales related to the time to empty a leaky bucket that are of interest here — see below for how this relates to measurement and estimation questions),
- obviously more complicated source models can be used, though it is questionable how far to go in this direction — any more complex model is no longer an upper bound for all traffic types, so it is necessary to be sure that it is being applied ‘correctly’; also a large part of the potential gain comes from using this simple model.

The calculations of resource or stream saturation are not complex. There is, however, a simple way to approximate them, making use of the theory of large deviations [13]. Large deviation theory is particularly useful for modelling ATM, since it relates to rare behaviour, which is exactly what is of interest with very small cell-loss ratios.

5.2 Effective bandwidths

This framework enables effective bandwidths to be defined. A number, α , the effective bandwidth of the connection, can be found such that n connections can be accepted and given quality of service constraints that are satisfied, if, and only if:

$$n\alpha \leq C \quad \text{..... (9)}$$

where the effective bandwidth is between the peak and mean rates of the connection:

$$\text{mean} \leq \alpha \leq \text{peak} \quad \text{..... (10)}$$

For example, if the quality or service constraints are given (equation (7)), then there is a maximum number of calls, $N_{\max}$, that can be accepted, in which case the effective bandwidth is simply given by $C/N_{\max}$. It should be noted that the effective bandwidth depends upon the grade of service target. Effective bandwidths apply quite generally, under a variety of models [14—15].

Example

For the AVA camera described in section 4.3, the mean rate is 2.74 Mbit/s. Table 1 shows the effect of different peak rates on the number of calls that can be multiplexed on to a 100 Mbit/s system, using a stream-based measure. Recall from section 4.3 that 9 Mbit/s, and 13 Mbit/s relate to having a buffer of size 1000, or 200 cells respectively in the first switch. If the contracted or policed peak rate is 9 Mbit/s, then 25 calls can be statistically multiplexed if the cell-loss rate is 10^{-2} , corresponding to an effective bandwidth (C/N) of 4 Mbit/s, 16 can be fitted on at 10^{-4} (6.3 Mbit/s effective bandwidth) and 11 if the peak rate is used, with corresponding figures for other peak rates. The figures are for illustration — 10^{-2} would not be an acceptable quality of service for this application.

Table 1 Number of calls that can be multiplexed on a 100 Mbit/s link.

	9 Mbit/s peak	13 Mbit/s	28 Mbit/s
10^{-2} CLR	25	20	11
10^{-4} CLR	16	11	3
no cell loss, peak allocation	11	7	3

The reason that useful multiplexing gain cannot be acquired at better qualities of service in this example is because the ratio of link rate to peak rate is too low, creating insufficient space for statistical multiplexing to work. The models are also very conservative in these regions, both the cell-scale model (where a worst-case assumption is being used) and the burst-scale, where on/off sources are assumed, and losses due to cell and burst-scale effects tend to blur. In fact, 14 AVA sources can be multiplexed together at a cell loss of 10^{-4} with a buffer of only 120 cells, as the simulation in Fig 9 shows, which plots the probability that the buffer size exceeds x cells on a logarithmic scale.

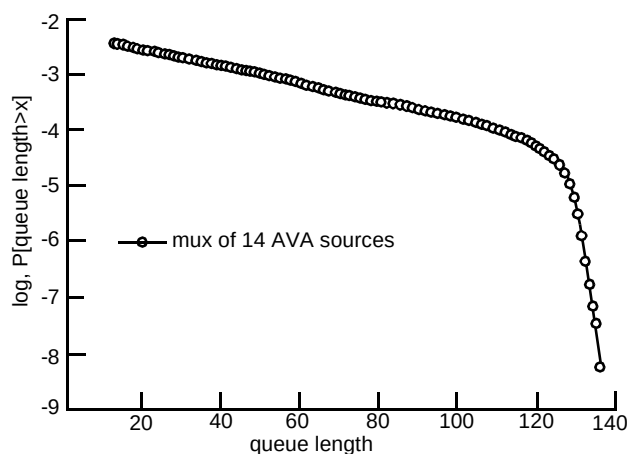


Fig 9 Logarithm of queue distribution function for 14 AVA sources.

A rule of thumb says that useful gains are only acquired when the link-to-peak ratio is at least ten, and ideally much higher.

Table 2 shows how things improve once the link rate is 600 Mbit/s. Notice too the effect of peak rates in the tables — for a given mean value, the lower the peak rate, the lower the effective bandwidth, hence the better for the network.

Table 2 Number of calls that can be fitted on a 600 Mbit/s link.

	9 Mbit/s peak	13 Mbit/s	28 Mbit/s
10^{-2} CLR	198	189	163
10^{-4} CLR	163	148	109
10^{-10} CLR	119	96	51
no cell loss, peak allocation	66	46	21

The multiplexing gain is defined as the ratio of the number of calls that can be multiplexed to the number that can be fitted on with peak rate allocation. In Table 1 the multiplexing gain is 2.3 (25/11) for a peak rate of 9 Mbit/s and 10^{-2} CLR, 3.7 for a peak rate of 28 Mbit/s. The corresponding gains in Table 2 are 3.0 and 7.8 respectively. In this example, even for a very tight cell-loss constraint, almost twice as many calls can be multiplexed as peak rate allocation on a 600 Mbit/s link. Figure 10 plots the multiplexing gain at a stringent quality of service — the height represents the multiplexing gain, while the capacity is the ratio of the link rate to the peak rate (equivalently the number of calls that can be fitted on using peak rate allocation). Recall that the burstiness is just the ratio of peak-to-mean rate, or $1/p$, and clearly the gain is bounded by the burstiness.

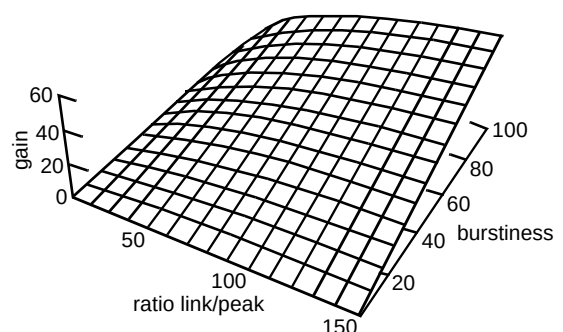


Fig 10 Multiplexing gain for a cell loss of 10^{-9} .

Figure 10 shows the state of the greatest gains when the capacity is large (i.e. the ratio link/peak is large), and the burstiness high. Conversely, the gains are small when the capacity is small, and the burstiness small. It should be

noted how the multiplexing gain increases almost linearly with burstiness.

The other factor is quality of service — the tighter the cell-loss constraint, the smaller the multiplexing gain. The QoS constraints have to be orders of magnitude different to have a big effect.

5.3 Utilisation and call blocking

The other option is to look at utilisation or load. Figure 11 is related to Fig 10, but now the utilisation rather than the multiplexing gain has been plotted. If n calls are accepted (those with 'activity' p), then the utilisation is just np/C . Again, Fig 11 relates to a cell-loss constraint of 10^{-9} .

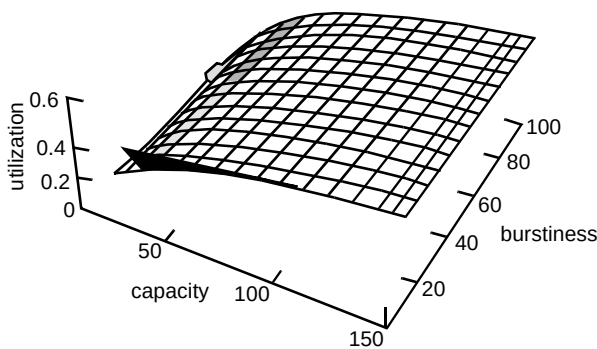


Fig 11 Utilisation against capacity and burstiness.

Notice now how the utilisation is very low for bursty sources which are large compared to the link rate (equivalently the capacity is small). In this case the utilisation can be as low as p , the activity of the source. However, this is effectively the best that can be done. To achieve higher utilisation either the sources have to be smoothed out, or more traffic has to be packed on to the link, which implies a much reduced grade of service.

In practice, even this utilisation will only be achieved in busy periods. A key determinant for real time services, and indeed for all connection-oriented services, is the connection or call blocking. This is the probability that a connection is refused service because there is not enough capacity. For telephony, this currently averages 1 in 1000 or better. Therefore the network has to be dimensioned to ensure such grade of service criteria are met. Once this is done, the expected utilisation over time will be lower.

There is an obvious trade-off between call blocking and cell loss; when a call arrives, it can either be rejected, or accepted with possible lost cells. Typically there is a call-blocking target which is several orders of magnitude lower than the cell-loss ratio, such as a call-loss of 1 in 10^{-3} and a cell-loss of 1 in 10^{-6} or better.

This trade-off is illustrated in Fig 12, which shows expected utilisation and expected cell loss for a link of size 50 where the peak rate is 1, $p = 0.2$, and calls (connections) arrive as a Poisson process with rate ν , chosen so that the offered load ($\lambda = p\nu$) is high. The graphs show what happens when up to N calls are accepted, with further call attempts being rejected until a call clears down. The load here is very high, so either the cell loss has to be preserved, which implies a very high blocking, and low utilisation, or cell loss has to be sacrificed to achieve better utilisation.

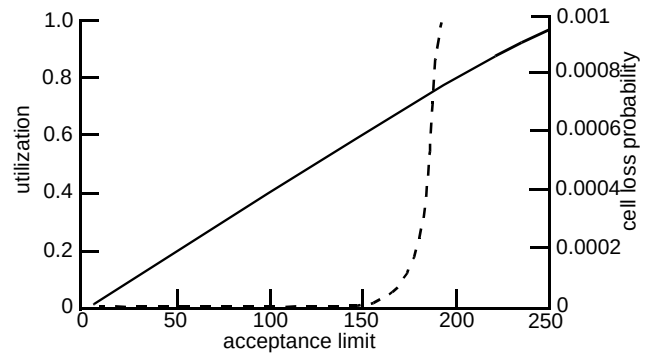


Fig 12 Utilisation versus cell loss.

5.4 Several types of traffic

What happens when there are several streams of traffic? Again, there are effective bandwidths associated with calls of type j , α_j , such that n_j calls of type j can be accepted and a given cell-loss constraint met, provided that:

$$\sum_j \alpha_j n_j \leq C \quad \text{..... (11)}$$

the generalisation of equation (9).

Just as before, the α depend upon the QoS target, but now they can also depend upon the traffic mix. This is because the actual acceptance region is the complement of a convex set. In fact, most regions are very nearly linear (see for example Griffiths [16]). There is really only a problem if traffic with very different peak rates is mixed, such as one source whose bandwidth is measured in megabits per second, and another measured in kilobits per second, when the acceptance region is quite concave. In that case, it is as though each source appears to the other as a constant bit rate source, thus reducing the effective capacity, and, as can be seen from Fig 10, the effective bandwidth depends upon the capacity. Even for such cases, constant effective bandwidths can still be used by defining the acceptance boundary as a tangent curve.

Constant effective bandwidths are required to facilitate resource allocation; it enables variable bit rate calls to be treated as if they are constant bit rate calls whose bandwidth

is just the effective bandwidth. Resource allocation consists of just adding up bandwidths.

5.5 Policing mean rates

The previous sections have shown that it is possible to statistically multiplex efficiently and still preserve target qualities of service. To do this it is necessary to know the required quality of service, the peak rate of the connection (and possible CDV tolerance), and, the most important thing, the mean rate. The first two quantities are straightforward, and expected — the quality of service may be a user-tailored quantity, or just a specified class, but either way it reflects what the user needs to obtain from the network in order for the users' applications to work properly. The peak rate is also limited by hardware constraints, and is a natural quantity to police. However, the mean causes problems.

It is not possible to police a true mean rate for statistical reasons — the mean can fluctuate on short time-scales, which means the policer has to be tolerant so as not to treat the user unfairly (e.g. use a large bucket depth); but this means that the network cannot respond quickly to contract violations, the time lag is too great, which could compromise the network. The ITU has defined a 'sustainable cell-rate' [5] with associated tolerance, which is like a short-term mean rate. This is policed by a leaky bucket, but with a smaller leak rate and larger depth than the peak-rate policer. The policed sustainable cell rate can be used as the mean rate in the calculations of effective bandwidths; but there are problems with this approach — the sustainable rate was originally defined for data sources which send information in fixed-length blocks or frames. Real time variable bit rate sources often use variable length frames. Moreover, the user has to be prepared to tolerate losing cells if they exceed this sustained contract; it is not certain that users will want this. But the real objection is a fundamental one — how will users know what their mean rate is? For a variable bit rate video codec, even with the same equipment, the information rate can vary according to the scene being viewed.

But why ask for information if you can do without it? If the network can estimate mean rates, then a lot of these problems are avoided. This is the solution proposed.

6. Adaptive call admission control (CAC)

6.1 Bayesian estimation

Only a mean rate need be estimated, or equivalently, the 'activity' parameter p , the mean-to-peak ratio. The focus will be on estimates of p . A simple estimate would just be to estimate this quantity directly — from the number

of sources currently active, or by counting cells in an interval of time and dividing by the peak number of cells that could be carried in the interval. (The appropriate time interval over which to measure is the time period of the associated leaky bucket.)

To be specific, recall that X_i is the bandwidth requirement of call i , and from now on capacity and bandwidth are normalised so that the peak of each call is 1. The naive estimate of the activity p is just:

$$p = \frac{\hat{S}_n}{n} \quad \dots\dots (12)$$

However, this will not do. The 'large deviation heuristic' [13] says that when rare events happen, they happen in the most likely way. In this context, with such an estimate, calls are lost through a combination of two (rare) events:

- an observed low number of active connections, which indicate that the activity may be underestimated — this means the acceptance of more calls than is appropriate,
- an unusually large number of active connections in the next time period, which exhaust the capacity.

The effects of estimation error need to be taken into account, and Bayesian decision theory provides a suitable framework [17]. Firstly, it is assumed that these 'beliefs' about the value of the activity p can be represented by a prior probability distribution $f(p)$. If there is really no information about the value of p , then uniform distribution can be used, which says that p is equally likely to be any number between 0 and 1. However, in practice, this is not likely to be the case; for instance, the value may be estimated from historical data, or from some knowledge of the application. Figure 13 shows two distributions that have the same mean but different variances, representing different beliefs; for the solid curve, there is much more certain information about the value of p .

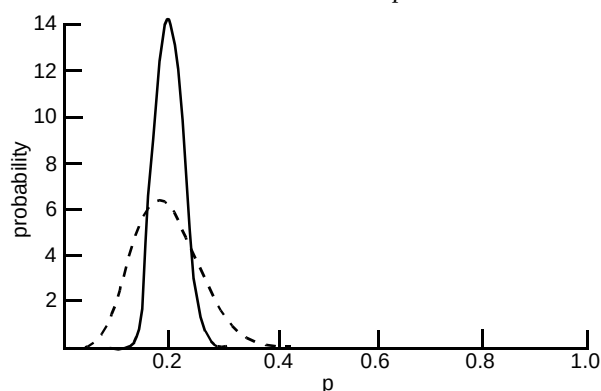


Fig 13 Different prior beliefs (with mean 0.2).

Now an observation of the load is given, $S_n = s$ say, which gives some information about the activity, and now a

simple application of Bayes theorem gives a posterior distribution, $f(p|s)$, which reflects the changing beliefs. Now a decision has to be made: should the next call be accepted? The posterior could be used to estimate the 'activity' p ; however, since the cell loss needs to be kept below the target value, a better approach is to require the expected cell loss to be less than the target value:

$$E_{f(p|s)} [\text{cell loss}] \leq 10^{-\gamma} \quad \text{..... (13)}$$

where the expectation is with respect to the posterior. This is straightforward to calculate, and gives an acceptance boundary in (n,s) space, the curve $s(n)$. A simple call admission procedure for a given value of n is to observe S_n , and when the next call arrives, accept it if the observed value of S is less than the boundary value $s(n)$. Recall that the peak has been normalised to be 1; hence the cell loss is zero, if n is less than C , so it is always accepted if $n \leq C$. The process is illustrated in Fig 14.

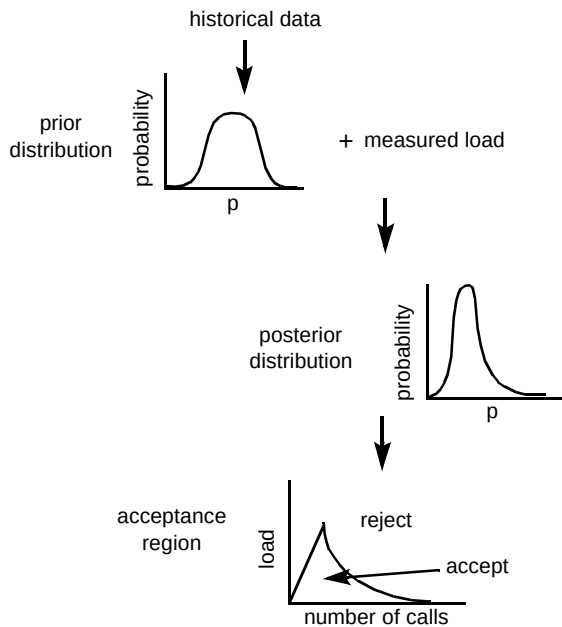


Fig 14 Adaptive call admission control.

An example is shown in Fig 15, with a 10^{-9} cell-loss target. Curves are shown for a uniform prior, and two other priors described earlier in Fig 13, which both have a mean of 0.2. Note that if the true value was known to be 0.2, then the acceptance curve would be a vertical line at $n = 119$. The vague (uniform) prior gives similar results to the curve where it is likely that the mean is around 0.2. If the prior information is very strong, then the curves can be quite different, as shown by the curve with 'strong' prior parameters.

The following points should be noted.

- Reading off from the $s(i)$ boundary curve a value of n for given value of s gives an effective bandwidth of C/n .

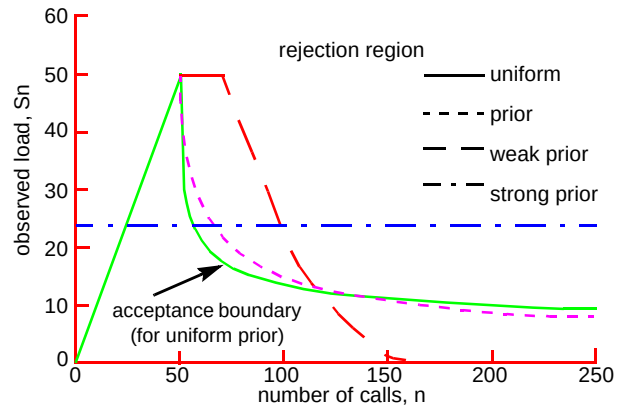


Fig 15 Acceptance boundaries.

- Calls with varying mean rates can be mixed safely — in other words there is no need to have calls with exactly the same activities since this procedure will overestimate the cell loss. (If there are calls whose activities are very different, then more can be multiplexed if the calls are split into different types, but at the cost of complexity.)
- The method could be used iteratively, using the posterior for stage n as the prior for $n+1$, as in a straightforward Bayesian analysis. It was decided not to do this, to avoid the estimation errors mentioned previously, and to allow the value of p to change over time.
- The same method can be applied to multiple call types, i.e call types which have different peak rates, or whose means are very different. For peak rates which are of the same order of magnitude, the effective bandwidths are simply added up to provide a simple approximation.
- Simulation results on real, heterogeneous traffic have shown that these methods work, and guarantee quality of service [12].

This approach is simple and conservative; however, it can be too conservative. A call acceptance control has been defined, rejecting calls if the observed load is too high, which means that the expected cell loss will be lower than the target (see section 5.3). If, on the other hand, there is a mass of call arrivals, and this leads to an extreme overload for a period of time, then the rate of admitting calls might be too large. These effects are caused by ignoring the arrival rate, which is discussed in the next section.

6.2 Load-based acceptance strategies

A defence strategy against high offered loads first described by Bean [18] is the backoff strategy — when a

call is rejected, accept no further calls for consideration until one current connection has ended.

This sort of strategy is likely to form part of a switch's defence mechanism anyway, but it is given explicitly here, and is assumed to be part of the CAC function.

Using the approach of section 6.1, an adaptive CAC strategy is now an acceptance curve, $s(n)$, where a call is accepted if $S_n \leq s(n)$ combined with the back-off strategy. The previous section took no account of the number of calls in progress, whereas observing this quantity provides information about the arrival rate — this is now taken into account.

Assume that calls or connections arrive as a Poisson process of rate ν , so the offered load is $\lambda = \nu p$. Using these assumptions, for any given acceptance curve $s(n)$, the probability $\pi(n)$ that there are n calls in progress can be calculated [19] (note that this will depend upon both λ and p). Hence the expected cell-loss ratio can be calculated, averaged over n .

The framework is now in place to implement a full decision-theoretic approach to adaptive call admission. The overall objective is to maximise the utilisation (carrying as many calls as possible) subject to the grade of service constraint. Suppose unit reward is received every time a call is carried and y is the penalty incurred for losing each cell. For example, $y = 10^9$ might be put to represent a cell-loss target of 10^{-9} . Let $M(n, p)$ be the expected number of cells lost if n and p are fixed. Then the objective is to choose the control function $s(n)$ to maximise the expected reward:

$$E_p \{E_n[np - yM(n, p)]\} \quad \dots (14)$$

where the inner expectation is obtained by summing over the states n with respect to the distribution $\pi(n)$, and the second by integrating with respect to the prior. This still depends on the offered traffic, so a prior distribution is used for the offered traffic, and hence integrated with respect to this to derive the acceptance curve.

Examples of the latter are given in Gibbens et al [19], where curves are produced which are similar to those of Fig 15. Indeed, in Gibbens et al [19] it is shown that almost nothing is lost by taking only a single observation and load measurements, compared to the case where the parameters of the distribution are known. In the example given, it is less than 3%.

6.3 Load-only measurements

The curves obtained using the methods of sections 6.1 or 6.2 are flat for a range of values, which suggests that just a

single acceptance curve of the form $s(n) = s$ could be used, which does not depend on n . This is equivalent to saying no account is taken of the calls in progress, setting aside this information. How should such a curve be chosen?

Equation (14) can be used with the restriction that only constant s curves are of interest. The actual value depends on the values of the prior chosen; however, little is lost by choosing the uniform prior. Even a value s can be chosen, based on a single value of p (a prior concentrated on a single point). For example, with $p = 0.2$, and a reasonable load (e.g. $\lambda = 50$), the value of s is 23.8, shown as the horizontal line in Fig 15.

This approach is remarkably robust and efficient — almost nothing is lost by using this approach compared to knowing p , and even if p differs from what it was thought to be, the expected cell loss is very close to the target value. To see this, consider what happens when p is actually larger than 0.2 (the only case that causes problems); then the observed load S will be higher than thought earlier on, and calls will be rejected before $n = 119$ is ever reached, with calls bumping into the boundary occasionally.

The benefits of this method are that it requires minimal information (just a measurement of current load), meets almost all of the achievable statistical gain, is robust, and can be extended to deal with several call types.

7. Tariffs, control and efficiency

This paper has presented an approach which makes use of a decomposition of time-scales, enabling an effective call admission control to be developed. The approach has assumed that connections have a relatively 'small' CDV tolerance, and has shown how this tolerance is related to buffer sizes in the switches. This is particularly appropriate for real time services, such as interactive multimedia connections.

What happens if a connection requires a large cell-delay variation? As has been shown, there is nothing sacrosanct about buffer sizes, and if the majority of connections wanted a tolerance of say 2.5 ms, then buffers of 1000 cells could be provided. However, this would mean that a voice or video connection starts to incur a large mean delay as it passes through the network, possibly requiring differentiation within a switch, allowing certain delay-sensitive services to go on some fast track through the switch. The problems increase if there are bursty data sources to be dealt with, which could transmit large bursts of back-to-back cells.

A related question concerns any variable bit rate sources. The larger the peak rate of the connection, the larger the effective bandwidth of the connection.

In both instances, the network could carry the source unaltered, or the user (or network) could shape the traffic,

decreasing the burst-length of the bursty data source, or decreasing the peak rate of the source. The course taken depends on the charging structure, and on the utilisation the network operator wishes to achieve. If bandwidth is relatively cheap, then the network's incentive to statistically multiplex decreases; bursty data sources can be carried by allocating them close to the peak rate. In the extreme case where bandwidth is totally free, there is no need to shape or alter traffic at all, but let each have its peak rate. At the other extreme, when bandwidth is very expensive or limited (as in the Internet or across international links), then there are pressures to shape traffic, or force users to offer to shape their traffic.

At the present time, even though relative transmission costs are falling, the absolute costs are still not trivial. At the same time, transmission costs are related to switch costs — an element of switch costs is the number of ports on the switch, and statistically multiplexing means efficient use of switch ports, not just transmission capacity. In certain cases the middle ground will be adopted — bandwidth costs are such that users could have a guaranteed quality of service, at one price, or opt for a 'best effort' service at a reduced price, relying only on a fair share. Both services could prevail, or the economics might push the market strongly in one direction.

8. Conclusions

This paper has attempted to show how connection admission control can be implemented, and has concentrated on open-loop control, which is particularly appropriate for real time services, and which polices connections at the entrance to the network.

The aim here is to require the minimum information from the user. For peak rate allocation, only two pieces of information are required — a peak rate, and a cell delay variation tolerance (CDV). The peak rate is used to determine whether there is enough bandwidth in the network to admit the connection, while the CDV determines whether there is enough buffer space in the switch. Buffer sizes are intimately bound up with the amount of CDV allowed, and it has been shown that this allows a 'natural' CDV tolerance to be defined.

To use the network more efficiently, variable bit rate calls can be statistically multiplexed. One way to do this is to require more of the user, such as some knowledge of the mean rate of the connection. Then an effective bandwidth can be calculated, and used in place of the peak rate to allocate bandwidth. But means are difficult to specify, and hard to police. Therefore on-line measurement is advocated as a way of discovering information about mean rate. The effects of estimation need to be understood, and accounted for — a mechanism has been provided for doing this, which is conservative, thus allowing statistical multiplexing while guaranteeing quality of service.

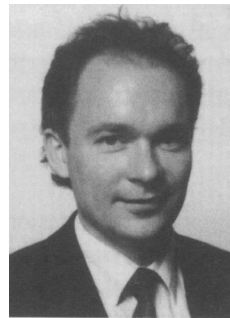
Acknowledgement

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References

- 1 Adams J L and Smith A J: 'The support of the available bit rate bearer capability using virtual source/destination concepts', *BT Technol J*, **13**, No 3, pp 67—79 (July 1995).
- 2 Shenker S: 'Fundamental design issues for the future Internet', to appear in *IEEE JSAC on Fundamentals of Networking* (1995).
- 3 Hui J Y: 'Resource allocation for broadband networks.', *IEEE JSAC*, **6**, pp 1598—1608 (1988).
- 4 Roberts J W: 'COST224 performance evaluation and design of multiservice networks', Final Report, Commission of the European Communities (October 1992).
- 5 ITU-T Draft Revised Recommendation I.371: 'Traffic control and congestion control in B-ISDN', Geneva (1995).
- 6 Cruz R L: 'A calculus for network delay part I', *Network elements in isolation*, *IEEE Trans Information Theory*, **37**, pp 114—131 (1991).
- 7 Gibbens R J and Kelly F P: Modelling cell delay variation in ATM networks, Technical Report, Report prepared for BT by Lyndewode Research Ltd (August 1993).
- 8 Crosby S, Leslie I and Key P: 'Cell delay variation and burst expansion in ATM networks', Results from a Practical Study Using Fairisle, submitted for publication (1995).
- 9 van den Berg J L and Resing J A C: 'The change of traffic characteristics in ATM networks', *COST242 TD(93) 035* (1993).
- 10 Kelly F P and Key P B: 'Dimensioning playout buffers for an ATM network', in *Proc 11th UK Teletraffic Symposium*, IEE, Savoy Place (1994).
- 11 Garrett M W and Willinger W: 'Analysis, Modelling and Generation of Self-Similar VBR Video traffic', *Sigcomm*, pp 269—280 (1994).
- 12 Griffiths T R and Key P B: 'Adaptive call admission control in ATM networks', in Laboutelle J and Roberts J W: 'The Fundamental Role of Teletraffic in the Evolution of Telecommunications Networks', *Proc 14th Int Teletraff Cong*, Elsevier Science (June 1994).
- 13 Bucklew J A: 'Large Deviation Techniques in Decision, Simulation and Estimation', Wiley, New York (1990).
- 14 Kelly F P: 'Effective bandwidths at multiclass queues', *Queueing Systems*, **9**, pp 5—15 (1991).

- 15 Elwadid A I and Mitra D: 'Effective bandwidths of general Markovian traffic sources and admission control of high speed networks', IEEE/ACM Trans Networking, 1 No 3, pp 329—43 (1993).
- 16 Griffiths T R: 'Analysis of a connection acceptance strategy for asynchronous transfer mode networks', IEEE Globecom, paper 505. 4, (1990).Lindgren B W: 'Statistical theory', Wiley (1976).
- 17 Lindgren B W: 'Statistical Theory', Wiley (1976).
- 18 Bean N G: 'Statistical multiplexing in broadband communication networks', PhD thesis, University of Cambridge (1993).
- 19 Gibbens R J, Kelly F P and Key P B: 'A decision-theoretic approach to call admission control in ATM networks', to appear in IEEE Journal on Selected Areas in Communications, Fundamentals of Networking (1995).



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